

VP Studio Suite

Operator Manual

Variable-data printing for the VPs printer family

TNR Developments

Revision 3.1 — September 2026 · suite version 0.144

Contents

1. The Suite at a Glance	4
2. Installation	4
Part I — Printing from Windows Programs	5
Quick Start: Printing from Word	5
The Placement Popup	5
Open in Studio	6
Copies, Multi-Page and High Volume	6
Part II — VP Studio	6
The Home Tab, Jobs and Recent Work	6
Settings	7
Designing a Piece	9
Variable Data and Tokens	10
The Data Page	10
Media Sizes	11
Pen Bands, Seams and the Optimizer	13
Proofing and Checking	13
Purging and Keep-Alive Jetting	13
The Print Page	13
The Machine on the Print Page	15
The Postage Page (Stamps.com add-on)	16
The Dealer Page	17
Ink Profiles	18
Scanning, Tags and Cards	19
Licensing	19
Updates and Rollback	20
Mock Mode — the Full Suite with No Printer	20
Part III — Tray Companion and Maintenance	21
The Tray Companion	21
Bench Maintenance	23
The Print Service	23
Diagnostics	23
Part IV — Troubleshooting	23
Appendix A — File Locations	24

Appendix B — Glossary

25

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1. The Suite at a Glance

The VP Studio Suite drives the VPs printer family two ways, through one shared background service that owns the printer:

- **The "VPs Studio" Windows printer** — print from Word, BarTender, Stamps.com, FlexMail, or anything else that prints. A placement popup lets you position the piece on the print heads before ink flies. Covered in **Part I**.
- **VP Studio** — the full layout and variable-data application: design pieces, connect databases, and run production jobs from its Print page. Covered in **Part II**.

Supporting both is the **tray companion** (printer settings, ink profiles, maintenance — **Part III**) and the **print service**, which starts with Windows and is the only thing that talks to the printer directly. Everything also runs in an interactive **mock mode** with no printer attached (see *Mock Mode*).

Inside VP Studio, five tabs run across the top: **Home** (the VP mark — recent and saved jobs), **Postage** (the Stamps.com counter — with the postage add-on, see *The Postage Page*), **Studio** (the design canvas), **Data** (records), and **Print** (production). Each tab shows a live line of context — piece size and object count, the loaded data file and record count, how many pieces are unprinted, the postage balance — and the right end of the bar shows the printer's live status at all times. Past the status sits the **dealer mark** — the logo (or well-set name) of the dealer who supports this machine; clicking it opens the Dealer page (see *The Dealer Page*).

2. Installation

Run **VpStudio-Setup.exe** (administrator rights required). If a previous version is installed you'll be asked what to do, and the first button tells the truth about the direction: **Reinstall** (same version), **Update to X** (newer), or **Go back to X** (older — for deliberate downgrades), plus Uninstall and Cancel. Updating keeps your settings. The installer automatically closes a running Studio, tray, or service before copying files.

Printer model page. Choose your printer. **VPs 2"** is the standard preset. Choosing **Custom** (or ticking *Show setup*) opens the configuration: total pens, number of heads, pens in each head, and transport direction, with a live schematic. With the printer plugged in, the page reads it and pre-fills the configuration for you to confirm. The dialog won't let you describe an impossible printer. This profile drives the band geometry and seam guides throughout the suite, so get it right here.

Ink profile page. Tick the inks you run. Each becomes an ink profile — the pen voltage, pulse and warming that ink needs — and the one you pick as *Profile a new job starts with* is the machine's **default profile** (see *Ink Profiles*). *Default* is always there. Scanning a cartridge later selects its profile automatically, and everything here can be changed from **File ■ Ink Profiles** afterwards. Re-running the installer never un-enables a profile you already have.

Media sizes page. Tick the stock this machine runs — envelopes, postcards, seed packets, plant stakes, labels — from the built-in trade-size catalog, or add your own width × height. These sizes appear in **every** program's paper list (Word included). Re-running the installer keeps the machine's existing list; everything can also be changed later from Studio or the tray (see *Media Sizes*).

Licensing page. The installer shows where this computer stands — a full license, a driver-only license, an expired one, or none — along with this PC's **Machine ID** (keys are issued against it), a **Request a Key** button, and an optional box to enter a key during the install. A **dealer code** can be entered here too — it brands the installation with your dealer's name, logo and contact details (see *The Dealer Page*). No key is needed to finish installing: everything runs in mock mode until one is added, and a key can be entered any time later under **Help ■ License** (see *Licensing*).

The installer also registers the **VPs Studio** printer in Windows (defaulted to Landscape), installs the printer's **USB driver** (a brand-new PC needs no other software), bundles the runtime it needs, starts the background service, and puts VP Studio on the desktop and the tray companion in Startup.

After installing: open VP Studio and check the right end of the tab bar — it should read *printer ready* (printer on) or *service running* (printer off). If it says the service is down, see Troubleshooting (Part IV).

Uninstalling. The uninstaller asks two deliberate questions after the usual confirmation, both defaulting to No: remove *everything* (settings, media sizes, profiles and logs), and *deactivate this PC's license*. A plain uninstall touches neither

— the license key and configuration survive for the next install. Saved jobs (.vpj files) are never touched either way.

Part I — Printing from Windows Programs

Quick Start: Printing from Word

1. Design your piece in Word (or any program). Landscape matches the transport direction and is the printer's default. The machine's own media sizes are in Word's paper list.
2. **File > Print**, choose the **VPs Studio** printer, set copies if you want, and print.
3. The **placement popup** appears with your piece and the print bands overlaid. Drag the bands (or click **Auto-place all**) so they cover your content.
4. Check the **Copies** box — it's pre-filled with what Word asked for, and *the number showing here is what prints*.
5. Click **Print Now**. Pieces buffer to the printer and print as the transport feeds media past the heads.

The Placement Popup

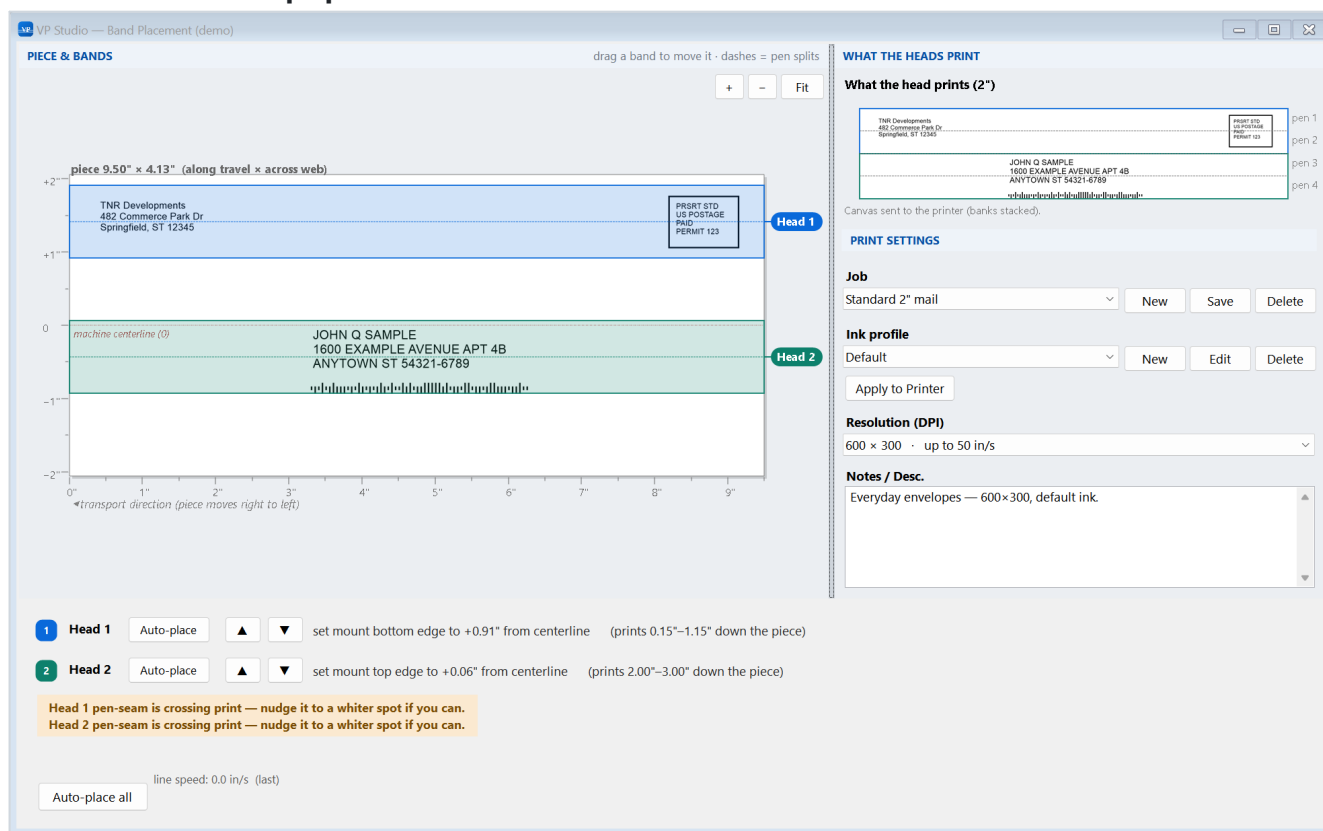


Figure 1 — The placement popup: both head bands with their stitch seams on the piece, mount-edge readouts per head, seam warnings, and the job's print settings.

The left pane shows your piece at product size with each pen band overlaid in color — the same head colors used everywhere in the suite — including every pen-to-pen stitch seam. The right pane shows the exact canvas the heads will print (bands stacked), with print settings below it.

Positioning bands. Drag a band directly, use its nudge buttons (0.05" steps), or click **Auto-place** on one band / **Auto-place all** — the optimizer covers all content, keeps the pen stitch seams out of barcodes first and text second, and maximizes breathing room. Each band's tag sits in the right margin with a leader line, so it never covers your print; while you drag, the tag shows the live physical mounting measurement — *the number you set the head hardware to*. Warnings appear as amber strips: a seam crossing print, or content falling outside all bands (it would be cut off). Any number of heads is supported; head rows carry numbered color chips matching their bands.

Copies. Pre-filled from the application's request. Type a different number and that's what prints — the operator always has the final say.

Multi-page jobs. A multi-page document is placed **once** — the layout applies to every page. Use the **Page N of M** browser to flip through pages; warnings recompute for the page you're viewing. Page browsing never changes the placement.

Speed readout. Live belt speed appears near Print Now, with a warning when the belt outruns the selected resolution's rated maximum — the countdown pauses while over-speed. If the printer can't report a fresh reading, the last known speed is shown marked **"(last)"** rather than pretending it's live.

Auto-print countdown. Tick *Auto-print in N sec* for walk-away operation: the popup places (using remembered or automatic placement) and prints itself when the countdown ends. Any band adjustment restarts the countdown. The setting is remembered.

Remembered placements. Pieces you've placed before reopen with your saved band positions, keyed by page size and content layout.

Other controls. Zoom (+/-/Fit, mouse wheel), **Jetting...** (keep-alive settings, see *Purging and Keep-Alive Jetting*), and **Cancel**, which cancels the Windows print job cleanly — including from the Windows print queue, at any stage. The popup opens even when the printer is off, so placement can be staged in advance; with no printer detected at all, the popup offers **mock mode** (see *Mock Mode*).

Open in Studio

The popup's **Open in Studio** button hands the job to VP Studio — artwork, page size, and your current band placement travel with it. You'll be asked:

- **Import** — the artwork lands as an editable full-page image; add live data fields on top.
- **Reference** — the artwork becomes a ghosted, non-printing tracing layer; rebuild the piece over it.

A **multi-page** document imports as *records*: each page becomes one record (a PAGE field), displayed and printed exactly like a multi-page PDF data source.

Copies, Multi-Page and High Volume

- Identical copies are stored on the printer **once** and re-printed N times — one transfer no matter the count.
- Very large jobs (label batches from BarTender, postage runs from Stamps.com) stream: pages decode as they print rather than loading up front.
- The **Windows print queue progress is honest**: it advances when pieces physically print (from the printer's own completion counter), not when they're merely queued.
- Multi-page + copies prints grouped by page (P1 ×N, then P2 ×N).

Part II — VP Studio

The Home Tab, Jobs and Recent Work

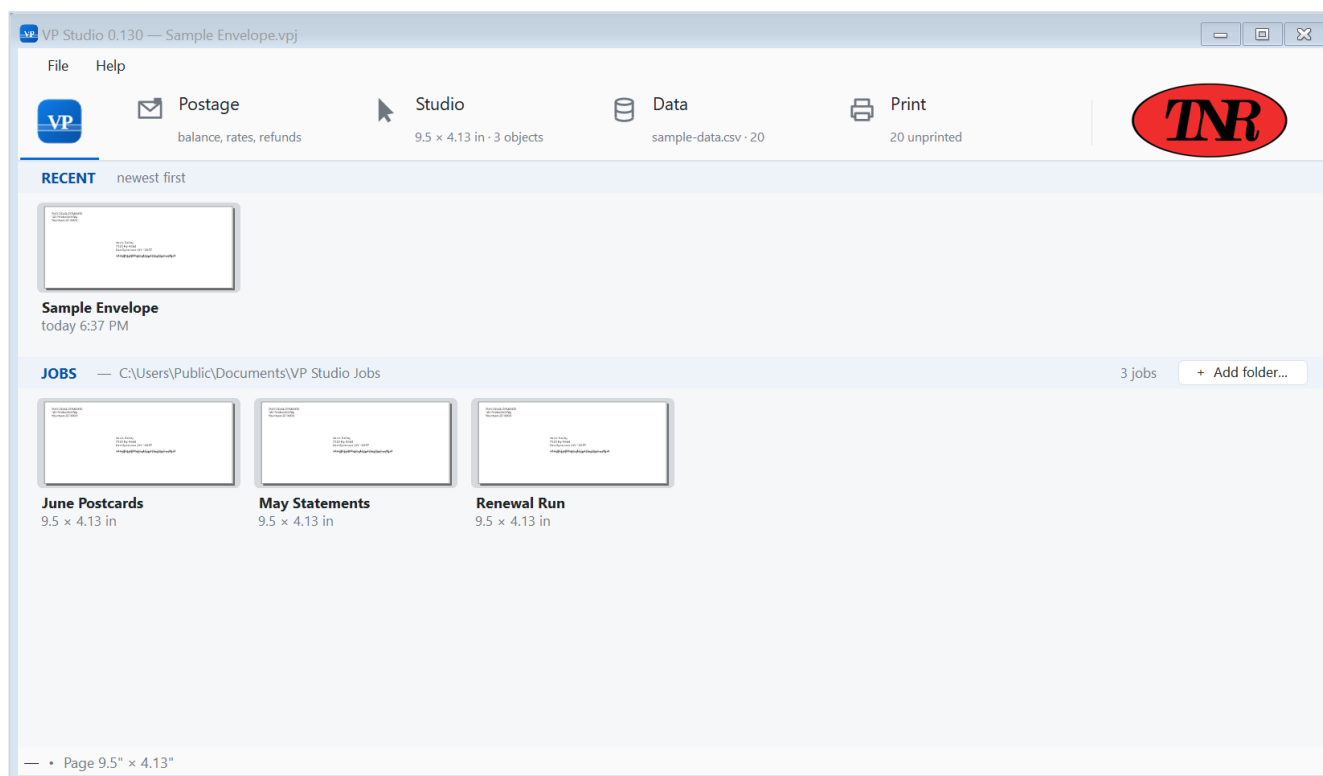


Figure 2 — The Home tab: recent jobs (newest left) and the jobs-folder sections, every card a live preview of the job's first piece.

The **VP mark** at the left of the tab bar is the **Home tab**. It shows your **recent jobs** in a strip (newest on the left), then a section for the **jobs folder** and any other folders you pin with **+ Add folder...** — as many as you like; each section scrolls on its own, snapping by row. Every job appears as a card with a real preview of its first printed piece. Click a card to open the job; right-click for **Open**, **Details...**, **Print...**, **Print a QR label...**, **Show in Explorer**, and (in the recent strip) **Remove from Recent**. A job that has a **job card** linked to it (see *Scanning, Tags and Cards*) shows a small **card TNR003** tag on its preview, and its right-click menu offers **Forget card** for each one. Details shows the piece size and object count, the data file and record count, **when it last printed**, the job's id and cards, and the file's dates and location.

Jobs save as **.vpj** files carrying the layout, data link, purge setup, output events, band positions, resolution, ink profile, and printed/reprint states; **.vpt** templates hold reusable layouts. New jobs save into the **jobs folder** by default — Documents\VP Studio Jobs unless you point it somewhere else in Settings — and you can still save anywhere. Recent jobs also live in the File menu.

By default Studio starts on a blank canvas. Settings changes that: pick your **startup page** (Home, Studio, Data, Print, or the **Kiosk** screen — see *Scanning, Tags and Cards*) and turn on **Automatically open the last job**. Start on Print with the last job open, and reprinting yesterday's job is just: launch, **Space, Space** (Space starts the run behind a confirm that answers to Space/Enter or Esc).

Settings

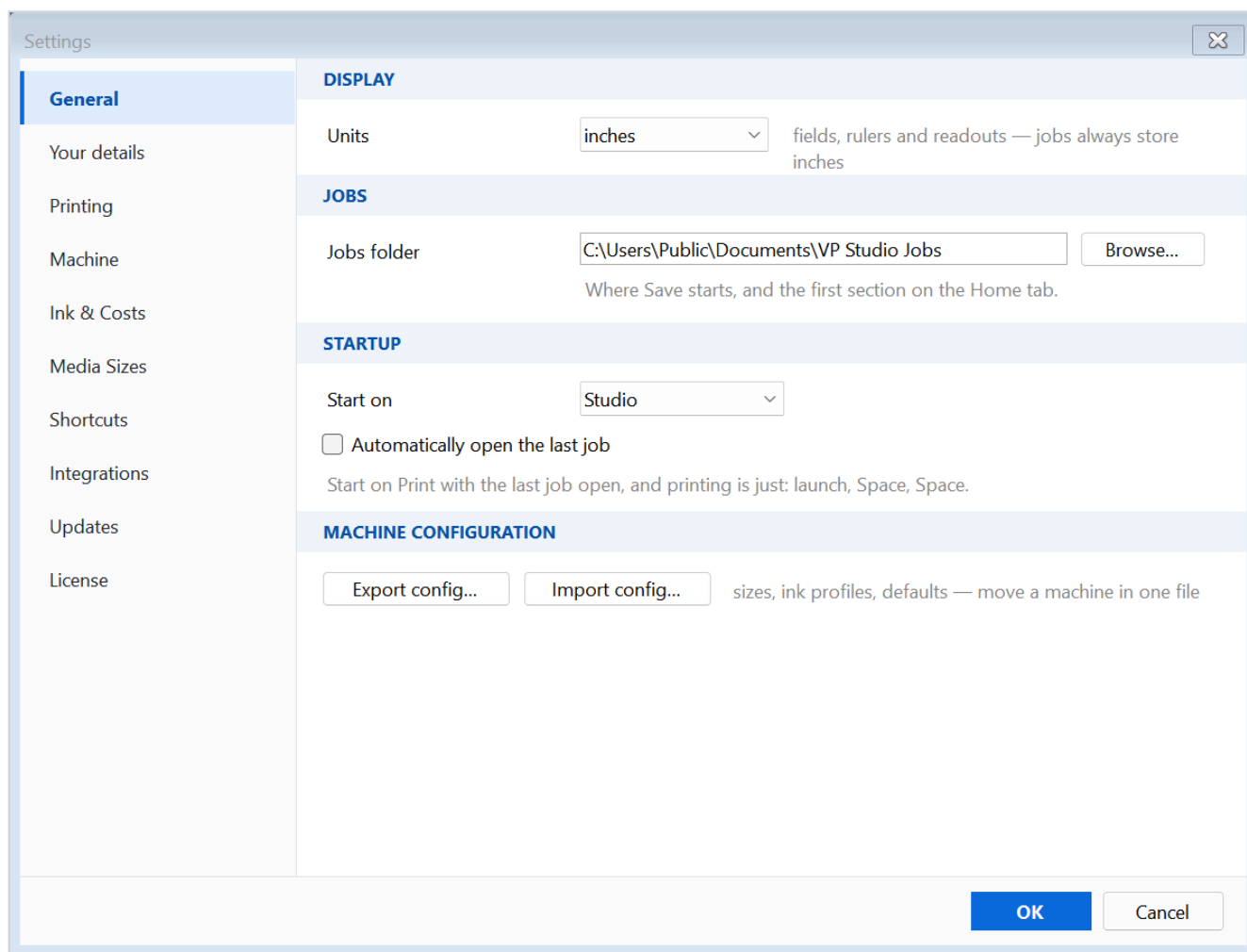


Figure 3 — File ■ Settings: General, Your details, Printing, Machine, Ink & Costs, Media Sizes, Shortcuts, Integrations, Scanner, Updates, License.

File ■ Settings opens one window with eleven pages:

- **General** — display units (**inches or millimeters** — rulers, fields and readouts; stored values are unchanged), the **jobs folder**, the startup page and open-last-job choices, and **machine-config export/import** (below).
- **Your details** — your shop's own details — name, contact, shipping address — used to pre-fill supply orders and license requests on the Dealer page (see *The Dealer Page*).
- **Printing** — the print buffer depth (how many pieces the printer holds ahead; default 4) and the machine's **batch separation defaults**: how THIS machine makes a gap between batches — run N blank pieces through, or fire a relay (set / clear / toggle / pulse for a time or distance, with an optional delay). "Break every N pieces" stays a per-job choice; jobs follow the machine default unless deliberately overridden (File ■ Batch Options).
- **Machine** — with the VP Link controller connected: how a run starts and stops the machine. The **ON sequence** and **OFF sequence**, four slots each — which channel, and how long to wait after it — plus a reset to the factory defaults for your model (see *The Machine on the Print Page*).
- **Ink & Costs** — the cost model: cartridge price, cartridge volume (mL), drop volume (pL), optional media cost per piece — plus the per-profile calibration the estimator has learned from real runs, with a reset. (Also reachable as File ■ Ink Cost.)
- **Media Sizes** — the machine's saved sizes at a glance, with the same Manage and Add buttons as everywhere else (see *Media Sizes*).
- **Shortcuts** — every keyboard shortcut, remappable: double-click an action, press the new key; conflicts are caught, and Reset all restores the factory map. Overrides live in the machine's settings, so a config export carries them.

- **Scanner** — the PC-side barcode/QR scanner (serial port or keyboard mode), the **countdown** a run card shows before printing, and **Speak scan events** — a spoken confirmation of what a scan did, for the operator at the feeder (see *Scanning, Tags and Cards*).
- **Integrations** — the Stamps.com connection: sign-in, the refund-policy switches, and diagnostics. The everyday postage controls live on the Postage page (see *The Postage Page*).
- **Updates** — the installed version, the newest published version (checked live), **Check now**, **Reinstall previous version**, and the daily-check toggle (see *Updates and Rollback*).
- **License** — the current license status and the license dialog (see *Licensing*).

Everything on these pages edits the same stored values as the standalone dialogs — either door works.

Machine-config export/import. Export writes ONE .vpconfig file carrying whatever you tick: media sizes, ink profiles, jetting configuration, printer-app settings, remembered placements, the print-app job list, Studio settings — and optionally the **jobs folder** itself (the .vpj files). Import is selective too: tick what to restore; every file replaced keeps a .bak copy alongside, and jobs are copied without ever overwriting existing files. Use it to set up a second shop, replace a PC, or keep a known-good backup before experiments. The tray menu has a quick "Export machine config..." as well.

Designing a Piece

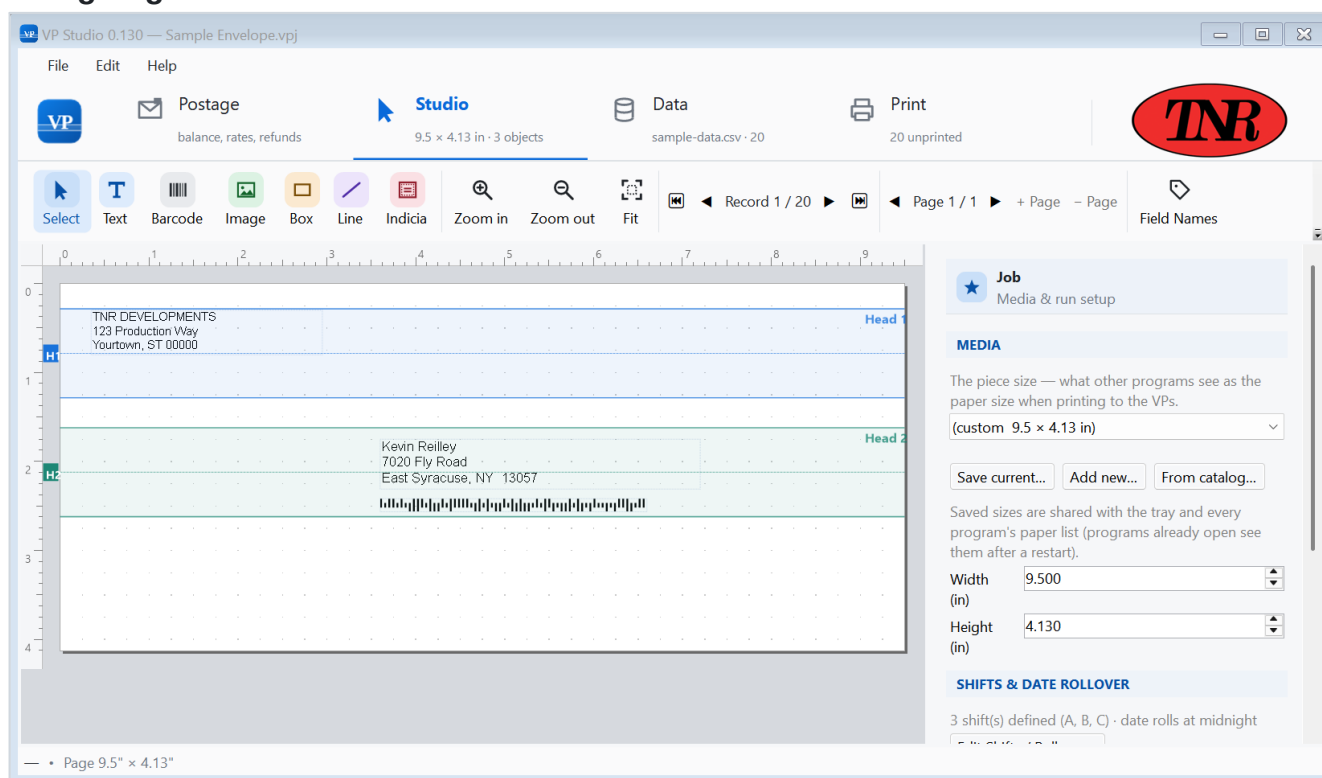


Figure 4 — The Studio tab: sample envelope with pen guides on — Head 1 and Head 2 bands with their true stitch-seam lines, live IMB barcode, and the Job panel's media setup.

The canvas has rulers (in your chosen units), a snap grid, zoom, and in-place text editing (double-click). Select with click, Ctrl-click, or marquee; align, nudge, duplicate, step & repeat, rotate, lock, and reorder from the toolbar/menus; 100 levels of undo. **Alt+drag** moves anything snap-free; **Alt+arrow** nudges exactly one print pixel (1/600"). Single-letter tool keys pick tools without leaving the canvas — **V** select, **T** text, **L** line, **I** image, **B** barcode, **X** box, **P** purge setup — and **F5–F8** jump between the four pages from anywhere (all remappable in Settings ■ Shortcuts).

Smart guides. While you move objects, dashed guide lines flash as their edges and centers line up with other objects, the page edges and center, and the pen bands' edges (when Pen Guides are showing) — and the object snaps to the line. A guide beats the grid; Alt bypasses everything.

Objects:

- **Text** — any installed font, size, bold/italic, alignment, per-line styling; effects: rotation, mirroring, outline-only, character spacing; *skip blank lines* keeps address blocks tight.
- **Barcodes** — **39 symbologies** rendered natively: USPS Intelligent Mail with live digit validation, POSTNET/PLANET, Code 128/39/93, Interleaved 2-of-5, EAN/UPC, GS1-128, **QR and Micro QR**, **DataMatrix (GS1/FNC1)**, **Aztec**, **PDF417**, MSI, Pharmacode, Codabar, Royal Mail, Australia Post, KIX, Japan Post and more — with bar-width reduction (shave/BWR), module width in mils, bearer bars, human-readable-text options, and quiet-zone intrusion warnings on the canvas.
- **Images** — fixed, or per-record by putting a «FIELD» token in the path. Mirror horizontally/vertically. Two print modes: **line art** (default — hard threshold, crisp indicia and logos) and **Photo** (dithered shading).
- **Boxes, lines and purge bars** (see *Purging and Keep-Alive Jetting*). Any image can be set **non-printing** as a tracing reference.

What you see is what fires: the print pass renders hard-edged and dot-exact.

Variable Data and Tokens

Insert into any text or barcode: «FIELD» database fields, counters (start/step), date/time formats including julian, «shift» codes (the shift table and the day-rollover time are editable under Shifts...), record numbers, and «prompt:Name» operator-entry fields (asked at print time; editable in User Fields).

Conditional printing — any object can print only when a field equals / contains / has any value. **Linked objects** — an object linked to the block above rides up when that block shrinks, so an IMB stays tight under a 3- or 4-line address.

The Data Page

VP Studio 0.130 — Sample Envelope.vpj

File Help

VP Postage balance, rates, refunds

Studio 9.5 x 4.13 in · 3 objects

Data sample-data.csv · 20

Print 20 unprinted

Load Data... Unload Data Use Sample Data Filter: Find: Find Next 20 records C:\Users\tjaec\AppData\Roaming\VPStudio\samples\sample-data.csv

RECORDS — sample-data.csv 20 records · showing all

	VEND_ID	VENDOR_NAME	PERSON	ADDRESS	CITY	STATE	ZIP	IN
▶	5	Alliance Fastener	Kevin Reilly	7020 Fly Road	East Syracuse	NY	13057	00
	6	Allied Electronics	Pat Hamilton	1157 Highland Ave, Suite 2	Cheshire	CT	06410	00
	7	American Electronic Res.	Debbie Regius	3505-A Cadillac Avenue	Costa Mesa	CA	92626	00
	9	Analog Devices	Karen Landers	3 Technology Way	Norwood	MA	02062	00
	11	ANCO Engineering Inc.	Joseph Kmec	217 Long Hill Cross Road	Shelton	CT	06484	00
	12	Anthem Electronics	Betty Anne Geier	61 Mattatuck Heights Road	Waterbury	CT	06705	00
	13	Arrow Electronics, Inc.	Debby Gilnack	860 North Main Street	Wallingford	CT	06492	00
	17	Avnet Electronics	Barry Dumphy	20 Sterling Drive, PO Box	Wallingford	CT	06492	00
	28	Bell Technology	Dawn Croteau	530 North Commercial St.	Manchester	NH	03101	00
	32	Cal-Greg Components	Katherine Mazoe	60 Alhambra Road	Warwick	RI	02886	00
	34	Capstone Electronics	Kathy Belanger	25 Upton Drive	Wilmington	MA	01887	00
	35	Carlo Gavazzi	Nancy Delbeny	10 Mupac Drive	Brockton	MA	02401	00
	36	CDW Computer Centers	Greg Eshoo	1020 East Lake Cook Road	Buffalo Grove	IL	60089	00
	37	Chronology Corp	Kevin Silver	8405 165th Ave NE	Redmond	WA	98052	00
	39	Circuit Solutions Inc.	Melissa Bacigal	1580 Lakeland Avenue	Bohemia	NY	11716	00
	41	Component Source, Inc.	Tom O'Connor	62 South Second Street	Deer Park	NY	11729	00
	48	Data Comm Warehouse	Mike Zator	1690 Oak Street	Lakewood	NJ	08701	00
	40	Datal Inc	Jennifer Bernier	11 Cabot Blvd	Mansfield	MA	02048	00

DATA SOURCE

File sample-data.csv

Records 20

Fields 8

Loaded today 6:51 PM

Reload Replace... Unload

FIELDS

VEND_ID 5 unused

VENDOR_N... Alliance Fastener unused

PERSON Kevin Reilly used x1

ADDRESS 7020 Fly Road used x1

CITY East Syracuse used x1

STATE NY used x1

ZIP 13057 used x1

IMB 0004061151275172... used x1

Sample = current record. Double-click a field to insert it into the selected text object.

THIS RECORD show

DATA QUALITY hide

all ZIP values are 5 digits

Page 9.5" x 4.13"

Figure 5 — The Data tab: the record grid is the print set; the side pane shows the source facts, every field with a live sample and usage badge, and the data-quality checks.

Load **CSV/TSV/delimited text**, **Excel .xlsx**, **dBase .dbf**, **Access .mdb/.accdb** (with table picker), or a **multi-page PDF** — each PDF page becomes a record whose artwork is the page itself. The grid is the print set: the filter defines it, search finds records, columns sort. Record identity is stable, so printed and reprint flags survive sorting, filtering, and reloads.

The **side pane** keeps the data honest. **Data source** shows the file's facts and can **Reload** it in place (same file, fresh contents — after the customer sends a corrected list), Replace it, or Unload. **Fields** lists every column with a live sample from the current record and a *used* ×*N* badge showing where it appears in the design — double-click a field to insert its «token» into the selected text object. **This record** shows the current record as a card. **Data quality** flags blank values in fields you actually use, duplicate addresses, and suspicious digit-length drift — with one-click "show" filters that pull the offending records into the grid.

Media Sizes

The suite keeps **one machine-wide list of media sizes**, shared by Studio, the tray, the installer, and — through the print support app — **every Windows program's paper list**. Add a size anywhere and it's everywhere (programs with an open paper list see it after a restart).

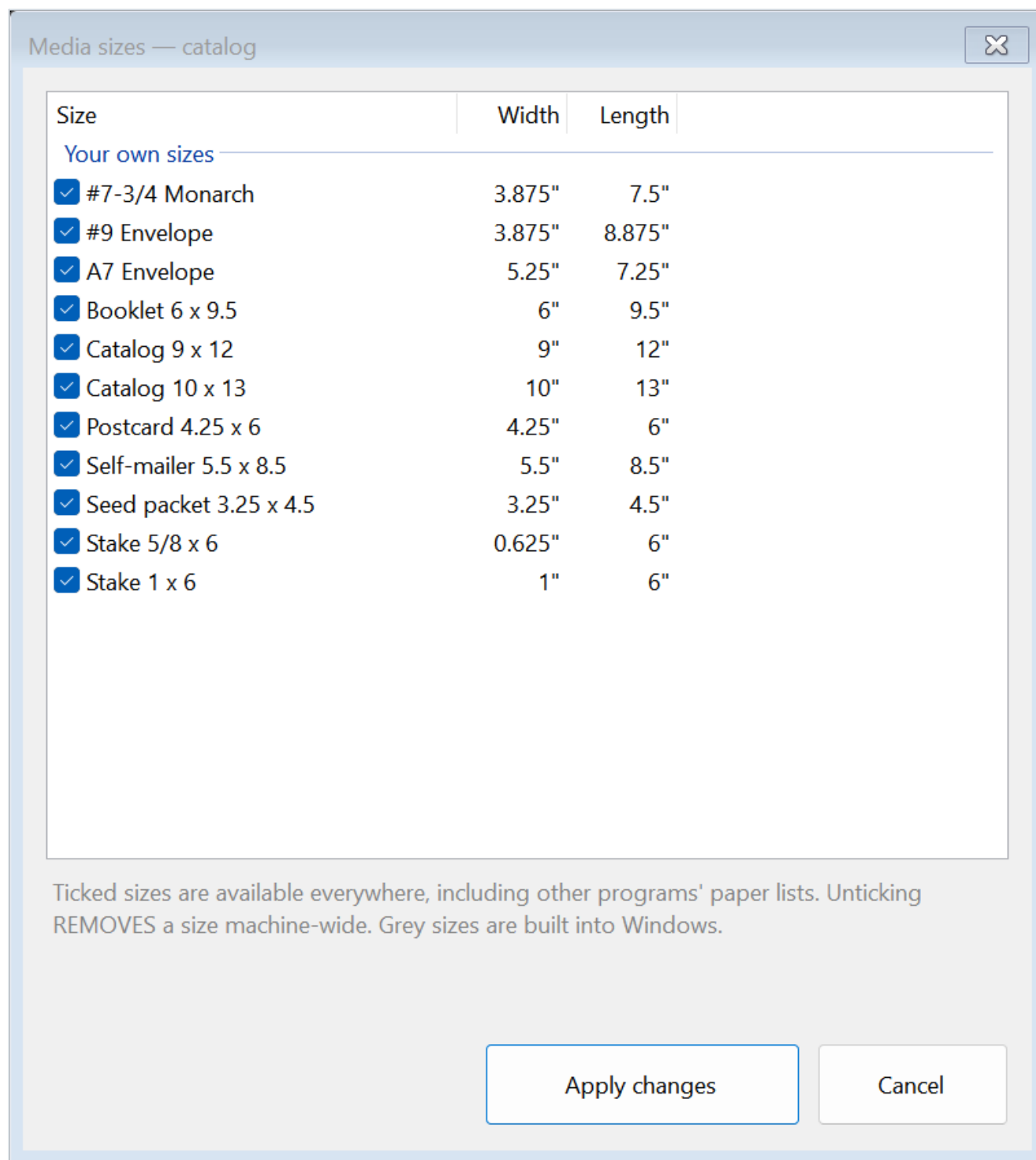


Figure 6 — The shared media catalog: tick to add a size machine-wide, untick to remove; your own sizes are removable in their own group.

In Studio: the Job panel's **Media** section has the saved-size dropdown (it names the size when your width/height match one), **Save current...**, **Add new...**, and **From catalog...**; the same list is in **File ■ Media Size** and Page Setup. The **catalog** holds ~55 named trade sizes — envelopes, postcards and mailers, seed packets, plant stakes and tags, labels and strips. The catalog manager is the same dialog everywhere: **tick to add a size machine-wide, untick to remove it**; your own hand-entered sizes are removable in their own group; standard Windows sizes are shown locked (Windows already provides them). Sizes run **long side = belt direction**.

Pen Bands, Seams and the Optimizer

Toggle **pen guides** to see the bands the heads print, drawn from your installed printer profile — including every pen-to-pen stitch seam (a 3-pen bank shows two seams). Click a band chip (H1, H2...) to select it, drag or arrow-key it, double-click a chip to auto-place that head, or type an exact top position in the properties panel; the readout lists the band range and all seam positions.

The **optimizer** places bands at true 1/600" using strict priorities: never cut content off a band; keep seams out of barcodes, then out of text; grow the weakest margin; keep a clear lead-in; center what remains. **Coverage warnings** hatch exactly the ink that would fall outside the bands, live while you drag.

Proofing and Checking

Proof mode flips through records print-accurately before you run. **Check Layout** (on demand from the Print page, or opt-in background checking) flags overflow, bad tokens, missing images, and blank pieces per record — results appear as flags in the Print grid. It scales to very large databases.

Purging and Keep-Alive Jetting

Thermal pens need to fire to stay healthy. Two tools, both previewed honestly on the canvas:

- **Purge bars** — dedicated bars that print at the start of a run (as a blank first page or on the first printed page) and/or every N pieces, overlaid on content or as inserted blank pieces. Patterns: solid, staircase, rotating segments, with darkness control. **Smart purging** inks only the nozzle rows that haven't printed real content within your piece/time threshold — partial-bar or whole-bar.
- **Random jetting** — scattered single-drop keep-alive dots injected service-side into every print (driver and Studio alike). Configure idle distance, scatter window, dot size, edge margins, and keep-away distance from real ink; preview on the canvas. Reachable from the Studio, the tray, and the driver popup's Jetting... button.

The Print Page

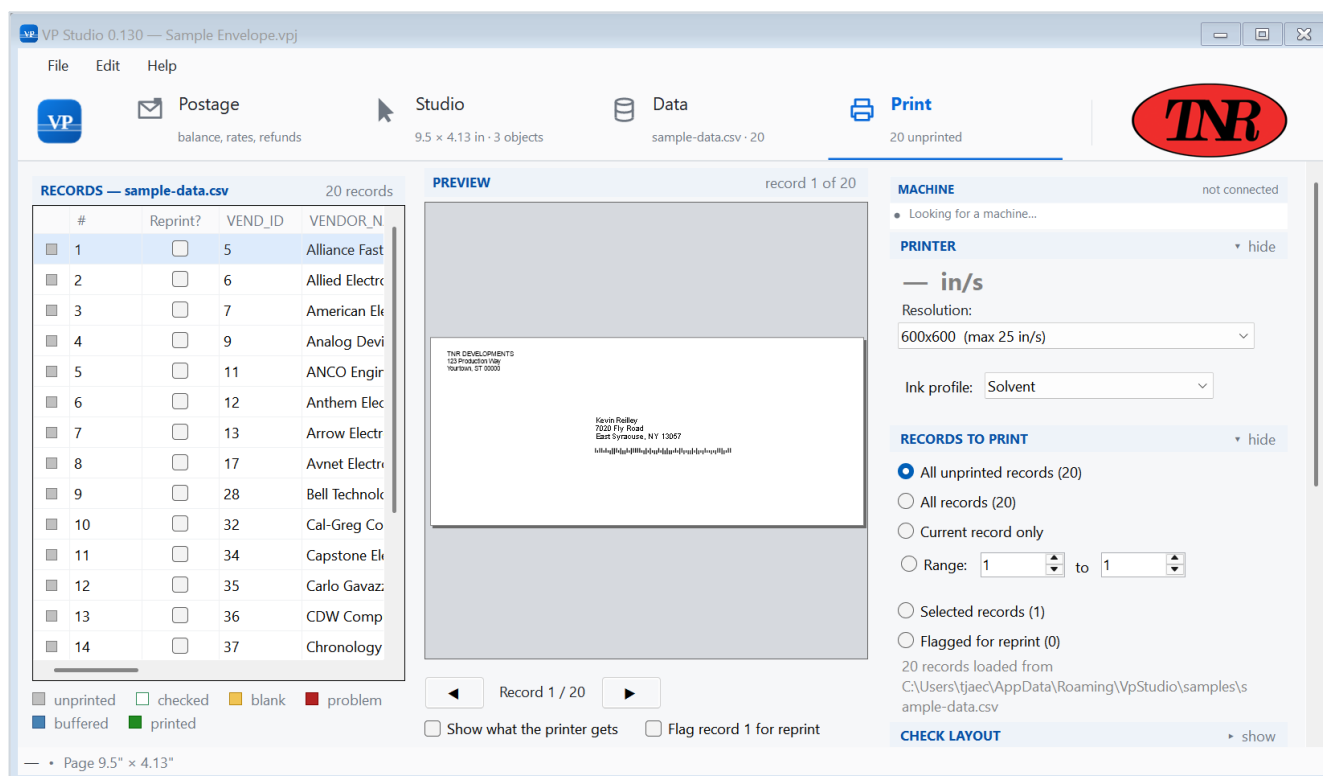


Figure 7 — The Print tab: a color flag per record, live preview, and the run controls — green means the printer itself confirmed the piece printed.

The record grid fills the screen, one color flag per record: **gray** unprinted · **outlined** checked-OK · **amber** blank · **red** problem · **blue** buffered in the printer · **green** printed — green means the *printer confirmed* the piece physically printed, from its completion counter. The page is organized in collapsible sections that remember whether you left them open, with a color-chip legend for the flags and live region readouts over the grid and preview.

Choosing what to print. All unprinted (the default — live counts on every radio), All, Current record, Range, Selected records (Shift/Ctrl-click in the grid), or Flagged for reprint. Set **copies per record** beside the range. Right-click records (or use the Edit menu) to Mark Printed/Unprinted, Flag/Clear Reprint, or jump to a record. Printed states save with the job and survive restarts — reprint-after-a-jam is: flag the survivors, print "Flagged for reprint."

Printer settings. The **belt speed gauge** reads live even while idle, so the transport can be dialed in before printing; it turns red showing the real speed when the belt exceeds the selected mode's maximum, and marks the reading "**(last)**" if the printer stops answering rather than freezing silently. The **resolution** list comes from the printer's own firmware modes with their rated speeds; the **ink profile** applies at job start. Both save with the job.

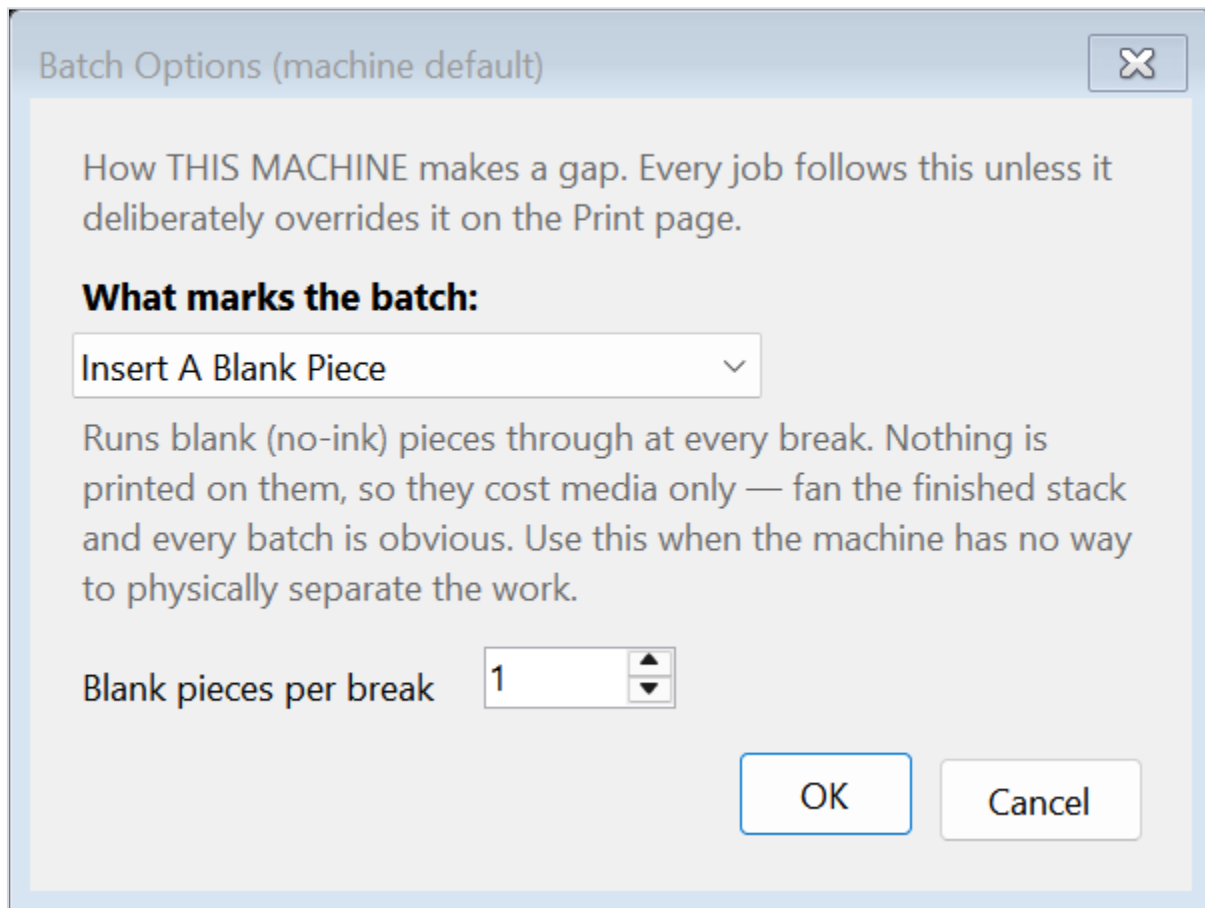


Figure 8 — Batch Options: how this machine makes a gap between batches — blank pieces or a relay event.

Batch separation. At every batch break the machine makes a gap the way its Settings say — blank pieces through the transport, or a relay firing a feeder stop, diverter, or beacon. The per-job part ("break every N pieces", or break on a data field change) lives with the job.

Output events. Fire the I/O-board relays on: run start/stop, low ink, blank page, blank purge page, or a database field trigger (equals / contains / changes — ideal for sort breaks). Actions: set, clear, toggle, or pulse (by milliseconds or inches of travel), with optional delays in ms, inches, or pieces. Events fire when a piece **physically prints**, not when it buffers, and are paced so back-to-back events stay distinct. Relay 3 exists only on two-headboard machines. Manual **Toggle Relay** buttons cover bench setup, and an abort mutes pending events.

Running. Print starts the job; the status box shows *printed / in buffer* live. **Abort** acts immediately: the printer's buffer is flushed, blue rows revert to gray, and the final count reflects only what physically printed. Printer errors surface as plain-language popups (including "Transport not moving" when the belt stopped). After each run a summary line shows pieces, purges, and **ink used** — volume, cost, and cost per thousand from the printer's real drop counters, which also self-calibrate the estimator per ink profile.

The Machine on the Print Page

With the VP Link controller board connected, a **Machine** block joins the Print page: the transport, feeder and vacuum as the controller sees them. Collapsed it is a live one-line summary; click to expand into per-channel detail. Two things it never hides: when the physical speed knob is holding a channel below what was asked for, **both** numbers are shown; and an alarm force-opens the block and cannot be collapsed away.

Manual control. Each channel can be started, stopped, and set to a speed — always within the physical knob's ceiling; the knob wins. Machine **Pause / Resume** buttons stop and restart the whole set without touching a job.

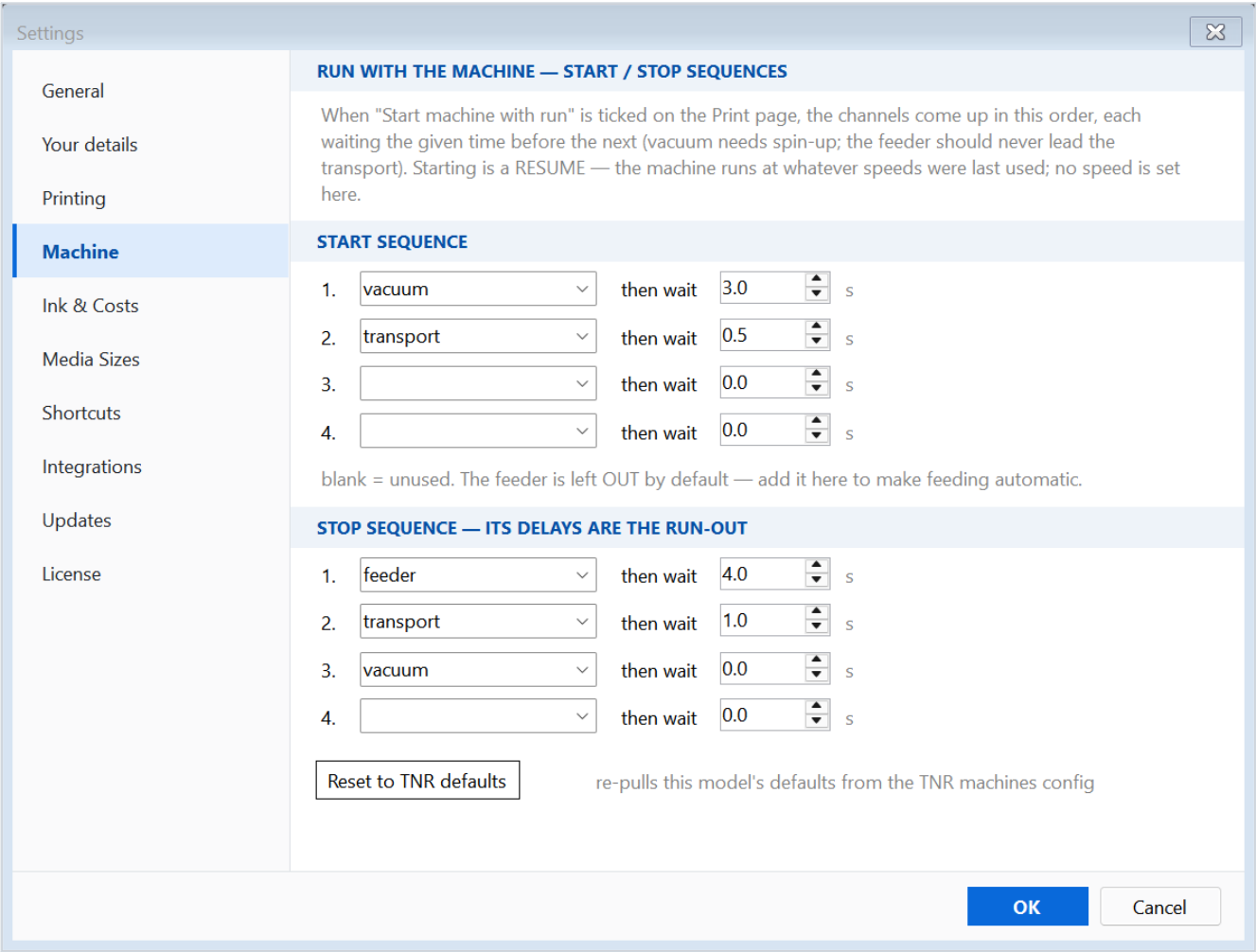


Figure — Settings ■ Machine: the ON and OFF sequences — which channel, and how long to wait after each.

Run automation (off by default). Two checkboxes beside the run controls: *Start machine with job* and *Stop machine after job*. Starting is really a **resume** — channels come back at their remembered speeds, in the configured ON order with a per-step delay (the vacuum gets its head start; the feeder never leads the transport). The OFF sequence *is* the run-out: it drains the last pieces through in order. **Pause** in Studio aborts the heads and runs the stop routine with the job left open, so one Resume later finishes the run with a single combined summary; **Abort** halts and latches. The order and timings live in **Settings ■ Machine**, seeded from the factory defaults for your machine model and editable per machine.

Faults, warnings and history. Drive faults stop the machine and are reported by name, straight from the drive manual. **Warnings** report without stopping — the block's header turns amber and the condition is marked (*still running*). Everything the machine does — faults, warnings, clears, halts, start-ups — is written to a date-stamped **machine event log** (Appendix A), and the **History** button opens a live, color-coded viewer.

The Postage Page (Stamps.com add-on)

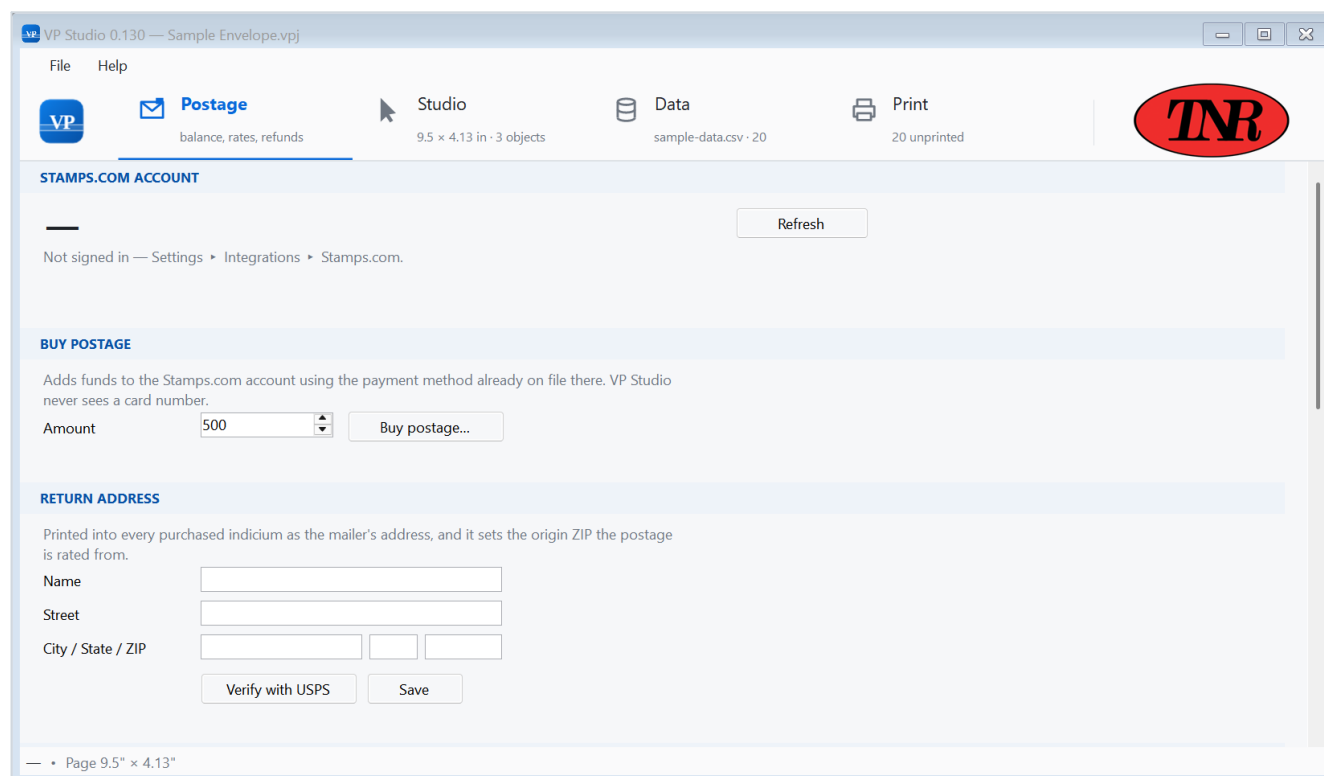


Figure — The Postage page: the live balance, buying postage, the return address, and what a piece costs.

With the postage add-on licensed, the **Postage** tab is the operator's counter — everything about the Stamps.com account touched during a print day, without opening a browser: the live **balance** (with Refresh), **Buy postage**, the **return address** (with *Verify with USPS*), and **What a piece costs** — pick a mail class and destination ZIP for a live rate. Postage paid for but never printed is claimed back automatically under the refund policy you chose. For what VP Studio can't do natively — payment methods, plan changes, printed-postage reports — the page hands you a signed link that lands on stamps.com already signed in. The account sign-in and the refund-policy switches live in Settings ■ Integrations.

The Dealer Page

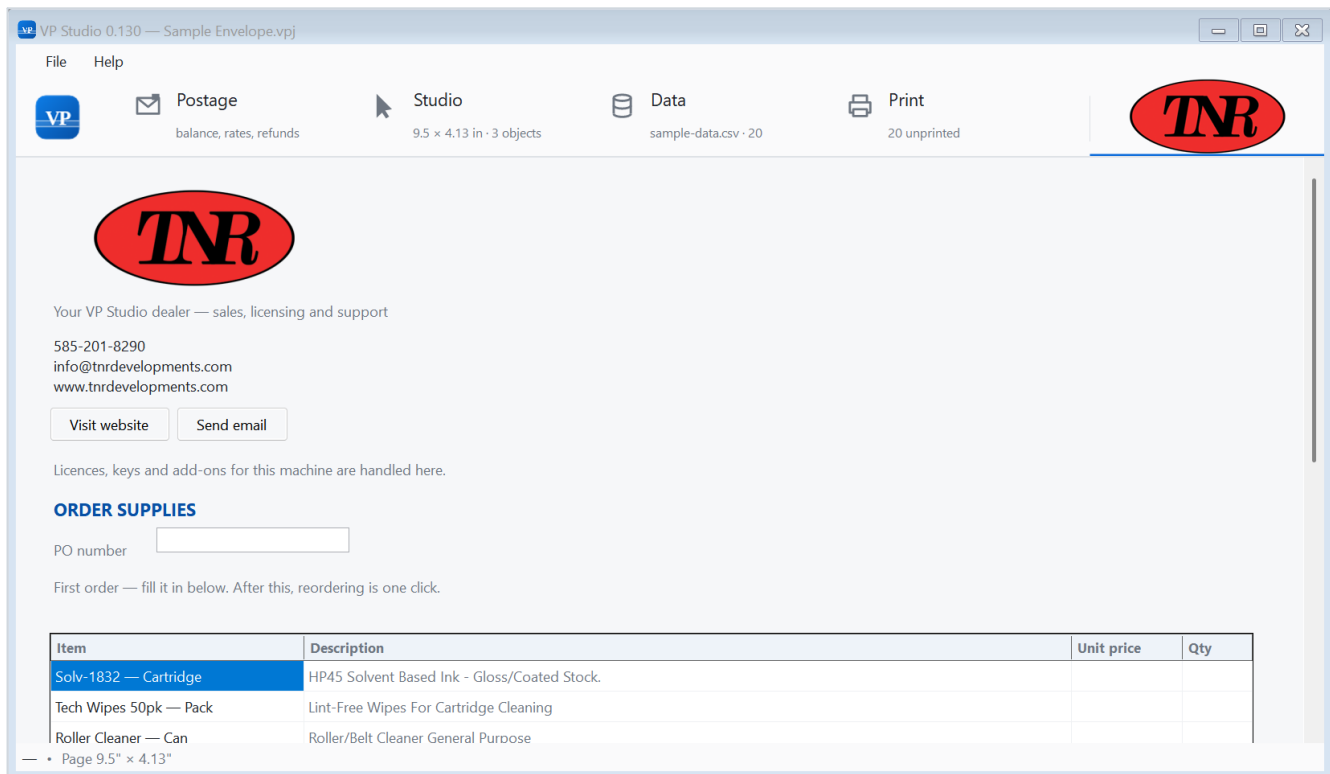


Figure — The Dealer page: who supports this machine, one-click supply reorders, and license requests.

The **dealer mark** at the right end of the tab bar — your dealer's logo, or their name well-set when no logo exists — opens the **Dealer page**: who supports this machine, with phone, email and website one click away. On an unbranded machine this is TNR's page.

Ordering supplies. Built for one click: **Reorder** repeats your last order, with exactly what it will send spelled out beside it; the item list underneath covers the case where the repeat is wrong. An order is a *request* to your dealer, who confirms, prices and bills the way you already do business. Your PO number is remembered after the first order.

License requests route through the same page — licenses are bought from your dealer — and your shop's details from Settings ■ Your details pre-fill both forms.

A dealer is applied with a **dealer code**: in the installer, or any time under Help ■ License. The code is verified against TNR's registry (live, when this PC is online), and the branding — name, logo, contact — keeps itself current in the background afterwards.

Ink Profiles

An ink profile is the pen electrical setup — cartridge voltage, fire pulse length, temperature and warming mode — matched to the ink you're running (e.g. *Default* 9.0 V, *Solv-1832* 8.4 V). **File ■ Ink Profiles** in Studio (or Printer Settings ■ Ink Profiles in the tray) edits them; they're shared suite-wide, selected per job on the Print page (and by the active tray Job for driver prints), and applied automatically when a run starts. **Apply to Printer** pushes one immediately for bench work.

TNR's factory profiles are listed in the dialog under *From TNR* — enable the ones you use; a profile you haven't edited follows TNR's updates, and one you have edited is yours from then on. One profile is the machine's **default (Set as default**, or right-click it): a new job starts on it, and so does a job saved before it had a profile of its own. A profile saved in a job always wins over the default. The installer's ink page sets the default on a fresh machine.

Ink tag / QR... in the dialog writes the selected profile's cartridge SKU to a tag or prints it as a code (below).

Scanning, Tags and Cards

VP Studio reads scan codes to save steps at the machine, and it writes them too. Every code is a short line of text — `VPINK:SOLV-1832, VPJOB:ABCD2345|Bag Labels` — carried either on an **NFC card** (tapped on the machine's reader) or in a printed **QR code** (read by the machine's reader or a PC scanner). Tap or scan, the result is the same, and a chime confirms every read.

Three readers, one result. The machine's own reader (its reads arrive through the print service), a scanner connected to the PC as a serial device, or an ordinary USB or Bluetooth keyboard-mode scanner — set up under **Settings ■ Scanner**. A code that isn't one of ours is ignored.

What a code does when it's read:

- **Ink cartridge (VPINK)** — selects that cartridge's ink profile for the whole printer, the current job and the tray's active Job, with a notice you can undo. A profile you haven't enabled yet is enabled by the scan. One ink per printer.
- **Job card (VPJOB)** — opens that job, found by identity in the jobs folder, your pinned folders and recent jobs — so a job that runs every week becomes a card on a hook instead of a file to find. If the job on screen has unsaved changes you're asked first, as always.
- **Run card (VPRUN)** — opens the job **and starts printing**. A notice counts down — *Starting "Bag Labels" — 130 of 250 records left in 5 s* — with **Cancel**, so a wrong card in the stack can be caught; the countdown length is in **Settings ■ Scanner** (0 starts at once). The card itself says whether the run **resumes where it left off** (the usual choice) or **starts from the beginning**. A card scanned while a run is live does nothing but tell you so. Load the stock and start the feeder as you normally would — the run waits for media like any other.
- **Supply item (VPSUP)** — opens the Dealer page's order form with that item added. And whenever the **order form is open**, any scan — ink or supply — adds that catalog item to the order and bumps its quantity on each repeat scan, instead of applying anything. On that screen the intent is to order.

Pre-written cards from TNR. Card packs come already written (`VPJOB:TNR001...`, or `VPRUN:TNR001...` for run cards), with the same code printed on them as a QR. Nothing is ever written on your end: tap or scan a new card and Studio asks **which job it belongs to** — the job on screen (saved first if it needs to be) or any saved job — then *Done — card TNR003 now opens "Bag Labels". Try it: tap the card again.* From then on the card just works. The job remembers its cards (you can hand one to each operator), the Home tab shows them, and moving a card to another job is a **Re-link...** on the notice or **Forget card** on the Home tab.

Tags & QR Codes (File menu). The tool for making codes yourself. Pick the type — ink cartridge, job or supply item — and the thing it stands for; for a job, tick *Start the run when this card is scanned* and choose resume or start-over. Then either **Write card** (present a blank NTAG card to the machine's reader; the result is verified), **Erase card...**, or use the **QR code**: copy it, save it as a PNG, or **Print as label.... Tag or QR for This Job...** opens the tool pointed at the job on screen.

Print as label... opens a small label window — a simplified Studio: the QR code, the job's name and the code text already placed on a business card (any stock size, landscape or portrait), the real pen bands with auto placement, *Add text, Duplicate, Delete*, then copies (or *until stopped*), ink profile, resolution, the machine panel and **Print. Save label...** keeps it as a normal job; **Open in Studio** hands it to the full toolset.

Kiosk mode (File ■ Kiosk Mode, F11). A calm full-screen page for a shop-floor PC: the printer's status, *Tap a card to start — or scan its QR code*, the job on screen and the last thing a scan did; while a run is live, the job name and its progress. Esc or *Exit kiosk* leaves it; Settings can make it the startup page.

A voice, if you want one. **Settings ■ Scanner ■ Speak scan events** has Studio say what a scan did — "Opening Bag Labels", "Starting Bag Labels in 5 seconds", "Solv-1832 ink applied" — through the Windows voice, for the operator standing at the feeder rather than the screen. Off by default.

Licensing

Two editions: **FULL** (driver + Studio) and **DRIVER-ONLY** (print from Windows programs; Studio designs and simulates but doesn't drive the printer). Alongside the edition key, **add-on keys** unlock optional modules — today, Stamps.com postage (see *The Postage Page*). Keys can be perpetual or expiring, and are issued against this PC's **Machine ID** — a short code shown in the license dialog that survives reinstalls (a Windows reinstall or system-drive swap changes it;

have the key re-issued then).

VP Studio License

Status: FULL license (driver + Studio), never expires.

Machine ID: (give this to whoever issues your key)

License key: - - - - - -

For licensing and support, contact:
 TNR Developments
 585-201-8290
 info@tnrdevelopments.com
 www.tnrdevelopments.com

Dealer code:

Supported by TNR Developments.

Figure 9 — The license dialog: status, this PC's Machine ID, key entry, and Request a Key.

Help ■ License shows the current status, the Machine ID (with copy), a segmented key-entry box (paste the whole key into any segment and it distributes itself), a **dealer code** box, and **Request a Key**, which opens tnrdevelopments.com with your Machine ID already on the clipboard. If your system was purchased through an authorized dealer, contact your dealer for licensing — a branded machine names them right on this dialog and on the Dealer page; otherwise contact TNR Developments — (585) 201-8290 — info@tnrdevelopments.com. Without a key, everything runs in mock mode (see *Mock Mode*) — no functionality is hidden, it just doesn't drive the printer.

Updates and Rollback

Once a day (if enabled), the suite checks for a newer published version and shows what's new — it **never installs anything by itself**. Update when offered, check on demand (Help ■ Check for updates, or Settings ■ Updates), or skip a version. Downloads are checksum-verified before they run. After updating through the updater, **Reinstall previous version** can take the machine back to the build the update replaced — verified the same way.

Mock Mode — the Full Suite with No Printer

When the service is running but **no printer is detected**, both print paths offer mock mode; the moment a real printer is plugged in, the offer disappears and the hardware takes over.

- **From Studio:** the Print page's status box shows *Printer not detected — [] Run in mock mode*. Check it and Print.
- **From Windows programs:** the placement popup carries a **[MOCK — no printer]** badge and note; Print Now starts the demo.

Either way the **trigger window** opens. The big **TRIGGER** button is the photo-eye: each click "prints" one buffered piece and displays it exactly as it would hit paper — the real pipeline output, watermarked MOCK on screen only. *Auto-trigger*

every N ms simulates a running belt. The window shows live Buffered/Printed counts, **relay LEDs** that follow your output events, and a timestamped event log. Everything downstream behaves exactly like hardware: rows green up as pieces "print," the Windows queue advances, relay events fire (virtually), buffering backpressure applies, and the cost calculator counts drops. Closing the window doesn't abort the run — Abort does.

Use it for training, job prep, demos, and testing output events — no ink, no media, no printer.

Part III — Tray Companion and Maintenance

The Tray Companion

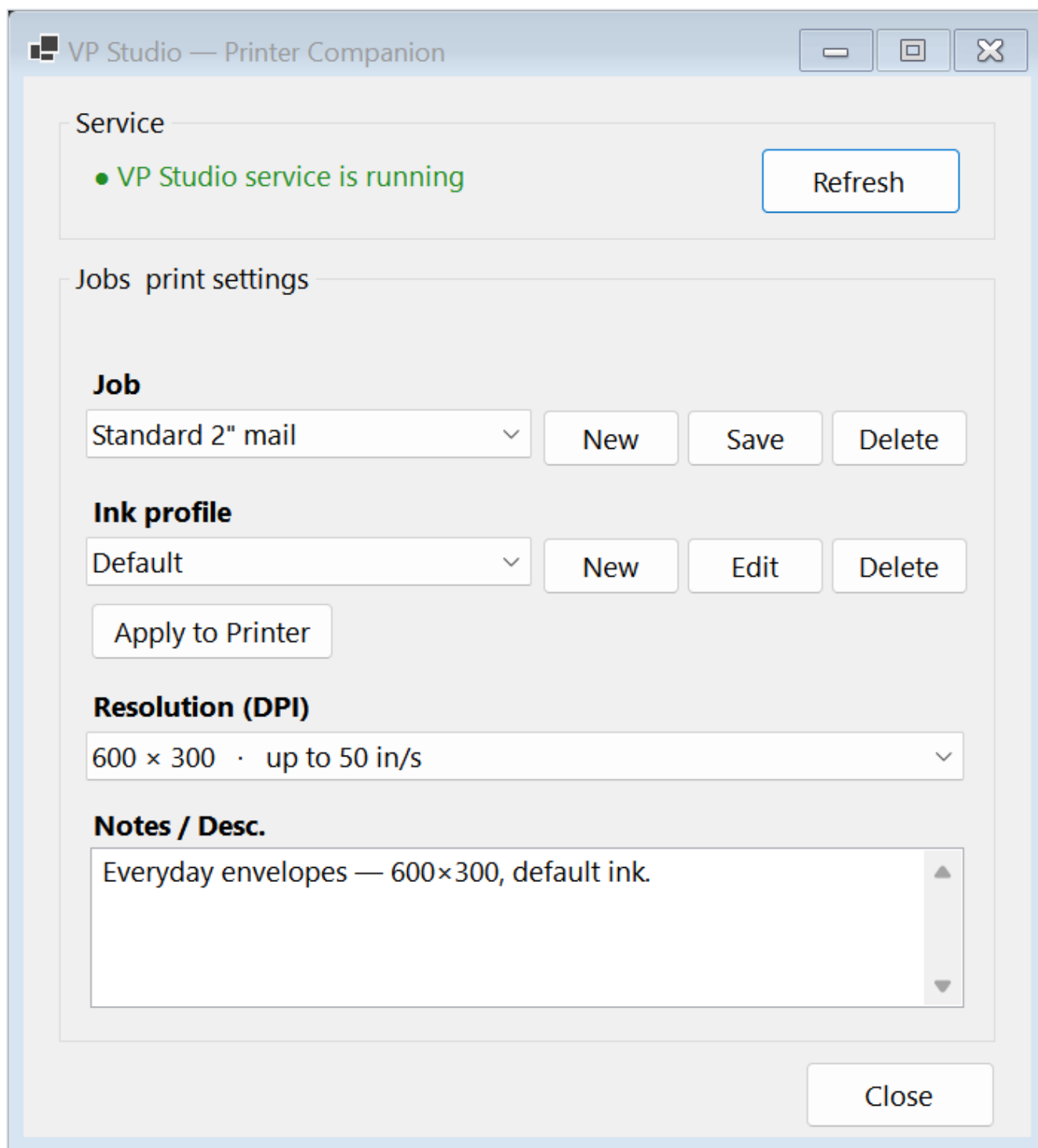


Figure 10 — The tray companion's settings window: Jobs (the active Job drives driver prints) and ink profiles.

Right-click the VPs tray icon (in the Windows notification area, near the clock — click the up arrow to show hidden icons if you don't see it). The menu opens with a live status line — printer ready, mock, or service down — and groups the everyday actions first:

- **Open VPs Studio Settings** — the settings window: **Jobs** (named bundles of resolution + ink profile; the *active* Job is applied to driver prints from Word etc.), **Ink Profiles** (edit voltage/pulse/warming; *Apply to Printer* pushes one now), and placement preferences.
- **Band placement preview** — a demo popup with a sample envelope for practicing placement.

- **Random jetting settings** — the keep-alive configuration (same file the whole suite uses).
- **Production history** — every run, driver and Studio, logged to a CSV: job, pages, duration, status.
- **Export machine config...** — the one-file config bundle (see *Settings*), quick form.
- **Config file editors** — direct access to the underlying JSON files for advanced tweaks.

Bench Maintenance

From the tray's maintenance functions (printer connected, transport not required):

- **Spit purge** (50/200/500 drops per nozzle) — put paper under the head first.
- **Pulse relay 1/2/3** — click-test the output wiring.
- **Software print trigger** — fires the next queued piece without a photo-eye, so full prints can be tested on a bench.

The Print Service

The service starts with Windows (no login needed), owns the printer exclusively, and serves both print paths — Studio and driver jobs can't fight over the hardware. It exposes a live status/diagnostics page on <http://localhost:60632/diag> (status, counters, rolling log). Log files and print dumps live under `C:\ProgramData\VpStudio` (Appendix A).

Diagnostics

Help ■ Diagnostics opens the suite's window into the hardware: a **Machine** section (the VP Link controller's live state), the complete **Drive fault list** from the drive manual — searchable, with each code's meaning, severity, and whether the Reset drive button can clear it — and **Print heads**. The footer names who to call: your dealer first, TNR underneath.

Part IV — Troubleshooting

Symptom	Likely cause	Fix
Tab bar says <i>service down</i>	Service not running	Reboot, or re-run the installer (Update). Check <code>C:\ProgramData\VpStudio\install.log</code> says <code>task_registered: True</code> .
<i>Printer not detected</i> / probe finds nothing	Printer off, USB unplugged, or driver missing	Power/replug; the installer stages the USB driver automatically — re-run it on a machine that never had one.
Placement popup never appears	Tray not running, or another placement is already open	Start "VP Studio Tray" from the Start menu; close the existing popup (one job places at a time).
Run fails: <i>Transport not moving</i>	Belt stopped, so the buffer never drained	Start the belt and print again — unprinted records are still marked unprinted.
Speed gauge red / run error at speed	Belt faster than the resolution's rated max	Slow the belt or pick a faster (lower-dpi) mode; the gauge shows the mode's max.
Speed shows "(last)"	The board stopped answering speed reads (usually during over-speed)	The number is the last real reading; it refreshes as soon as the board answers again.
A size added in Studio isn't in Word	Word's paper list was already open	Restart Word — paper lists are read at program start.
Print is faint / wrong darkness	Wrong ink profile for the ink	Select the matching profile (e.g. Solvent) on the Print page or active tray Job.
Blurry print	Media gap, belt speed vs resolution, ink/media pairing	The dump PNGs in <code>C:\ProgramData\VpStudio\dump</code> are dot-exact — if the dump is sharp, the cause is physical.

Symptom	Likely cause	Fix
Rows stuck blue after a problem	Pieces flushed at abort/error	They revert to gray automatically; if a run died hard, use right-click > Mark Unprinted.
Studio designs but won't print	Driver-only license, or no license	Help ■ License shows the status; Studio runs mock-only without a FULL license.
Installer reports files in use	An app survived auto-close	Close VP Studio/tray manually and re-run setup.
Machine block header amber	The drive reported a warning — the machine keeps running	The warning names itself in the block; History shows when it appeared and cleared. Warnings never stop a run.
Machine stopped mid-run with a named fault	The drive faulted (or E-STOP)	Fix the cause, Reset drive, resume. The machine event log has the full timeline (Help ■ Diagnostics lists every code).
Dealer code "not recognised"	Code mistyped, or this PC is offline	Check the code and the connection — codes are verified live with TNR's registry.
Something else	—	C:\ProgramData\VpStudio\service.log (rolling service log) and /diag tell the story; send both with a report.

Appendix A — File Locations

What	Where
Program	C:\Program Files\TNR Developments\VP Studio (bin = service, tray, studio, driver)
Printer profile (install wizard)	C:\ProgramData\VpStudio\printer-config.json
Media sizes (shared, machine-wide)	C:\ProgramData\VpStudio\custom-sizes.json
Ink profiles / jetting / placements / tray jobs	C:\ProgramData\VpStudio*.json
License key	C:\ProgramData\VpStudio\license.key
Production history + lifetime counter	C:\ProgramData\VpStudio\history.csv, lifetime.json
Machine event log (faults, warnings, halts)	C:\ProgramData\VpStudio\machine-events.log (rolls to .1)
Machine run-sequence defaults per model	C:\ProgramData\VpStudio\machines.json
Service + install logs	C:\ProgramData\VpStudio\service.log, install.log
Print dumps (dot-exact PNGs)	C:\ProgramData\VpStudio\dump
Studio app settings	%AppData%\VpStudio\studio-settings.json
Home-tab job previews (cache)	%LocalAppData%\VpStudio\thumbs
Jobs / templates	the jobs folder (default Documents\VP Studio Jobs) or wherever you save them (.vpj / .vpt)
Machine-config bundles	wherever you export them (.vpconfig)

Appendix B — Glossary

Band — one contiguous strip of print, produced by a bank of pens, positioned physically on the machine and mirrored by the on-screen guides. **Pen** — one HP45 cartridge (½" of nozzles). **Stitch seam** — the boundary line where two pens in a band meet; keep barcodes off it. **Piece** — one printed product (one record, or one purge page). **Buffered** — accepted into the printer's memory, not yet printed. **Printed (green)** — confirmed on paper by the printer's counter. **Purge bar** — sacrificial ink that keeps idle nozzles alive. **Random jetting** — scattered invisible keep-alive drops. **Ink profile** — pen voltage/pulse/warming preset per ink type. **Machine ID** — the short code a license key is issued against. **Jobs folder** — the default home for saved jobs and the first section of the Home tab. **Mock mode** — the whole suite running against a simulated printer with a manual trigger. **Fault** — a drive condition that stops the machine; Reset drive clears it once the cause is gone. **Warning** — a drive condition reported by name while the machine keeps running. **Run sequence** — the ordered ON/OFF channel list, with per-step delays, that starts and drains the machine around a job. **Dealer code** — the short code that brands an installation with its supporting dealer.